

PROBLEM SET 4—SUPPLEMENTARY MATERIAL FOR OFFICE HOURS AND TUTORIALS

ECON 101—INTRODUCTION TO MICROECONOMICS

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This week, the students will need to submit Assignment 5, part I and part II. Part I is straightforward multiple choice exercises, but Part II is more involved. They will need to fill out a table, similar to the one I did in the lecture, about the distortions in surplus generated after a tax or subsidy are imposed. I suspect that many students will need some guidance on this during individual office hours this week. Also, in this week's tutorial the GTAs will go over part II of the assignment.

To help with the preparation, I have prepared this document. I covered all this material in Lecture 6, part 2, so the students already have access to it in the supplementary material for the video lecture. However, there is a slight difference in notation here from the one that I used in the lecture. Thus, since I wouldn't like to confuse the students, please keep this document to yourselves, and only use it in support for your preparation of office hours/tutorials.

Recall that when a tax is introduced into a market by the government, we must distinguish between the price paid by consumers, P^d , and the price received by sellers, P^s .

Suppose that the demand and supply equations are given by

$$Q = A - aP^d \quad (1)$$

$$Q = B + bP^s \quad (2)$$

where A , a and b are positive numbers and B can be any real number.

Please note that compared, to lecture 6, the difference in notation in there is with respect to the demand and supply parameters. I am sorry about this discrepancy, but I prepared most of this document for the course in Fall 2019. For Fall 2020 I changed the notation for linear demand and supply schedules and I kept going with it. Table 1 gives the conversion between the parameters in here, and the parameters in the lecture.

Parameter in this note	Parameter in the Lecture
A	a
a	b
B	c
b	d

Table 1: Conversion table: Parameters in this document vs Parameters in the Lecture

Thus, in the lecture, the demand and supply equations were:

$$Q = a - bP^d \quad (3)$$

$$Q = c + dP^s \quad (4)$$

So, please adjust accordingly when talking to students. In what follows, I use (1) and (2) in the calculations. Let the per-unit tax be given by T . Then we know that

$$T = P^d - P^s \quad (5)$$

1. Using equations (1), (2) and (5), find the algebraic expressions for:

- Equilibrium price without tax (the so called efficient market price). Denote this price by P^e
- Equilibrium quantity without tax (the efficient market quantity). Denote this quantity by Q^e
- The price paid by consumers after a positive tax $T > 0$ is imposed.

- (d) The price received by sellers after a positive tax $T > 0$ is imposed.
- (e) The after-tax equilibrium quantity. Denote this quantity by Q^T
- (f) The government revenue from the tax.
- (g) The tax incidence, that is calculate the ratios $(P^d - P^e)/T$ and $(P^e - P^s)/T$.
- (h) The dead-weight loss generated by the tax.

Solution

- (a) $P^e = \frac{A-B}{a+b}$
- (b) $Q^e = \frac{aB+Ab}{a+b}$
- (c) $P^d = \frac{A+bT-B}{a+b}$
- (d) $P^s = \frac{-aT+A-B}{a+b}$
- (e) $Q^T = \frac{-abT+aB+Ab}{a+b}$
- (f) Government revenue is $\frac{T(a(B-bT)+Ab)}{a+b}$
- (g) Consumer incidence is $\frac{b}{a+b}$ and producer incidence is $\frac{a}{a+b}$
- (h) $DWL = \frac{abT^2}{2(a+b)}$

2. Now consider the case of a **per-unit subsidy**, $S > 0$. In this case we will have

$$S = P^s - P^d. \quad (6)$$

Repeat the same calculations as in part 1, that is, find the algebraic expressions for:

- (a) The price paid by consumers after the positive subsidy is imposed.
- (b) The price received by sellers after the positive subsidy is imposed.
- (c) the after-subsidy equilibrium quantity. Denote this quantity by Q^S .
- (d) The amount spent by the government to fund the tax.
- (e) The subsidy incidence, that is calculate the ratios $(P^e - P^d)/S$ and $(P^s - P^e)/S$.
- (f) The dead-weight loss generated by the subsidy.

Solution

- (a) $P^d = \frac{A-bS-B}{a+b}$
- (b) $P^s = \frac{aS+A-B}{a+b}$
- (c) $Q^S = \frac{abS+aB+Ab}{a+b}$
- (d) Government expenditure is $\frac{S(a(bS+B)+Ab)}{a+b}$
- (e) Consumer incidence is $\frac{b}{a+b}$ and producer incidence is $\frac{a}{a+b}$, similar to the case of a tax.
- (f) $DWL = \frac{abS^2}{2(a+b)}$

3. Consider the expressions that you obtained in parts 1 and 2 and observe them carefully. They depend on the *distortion generated by the tax/subsidy* which is the after tax/subsidy equilibrium quantity. What happens to the quantities you calculated when:

- (a) We have a perfectly inelastic an inelastic demand and a regularly sloped supply (elastic), that is, when $a = 0$ and $b > 0$.
- (b) We have a perfectly inelastic supply and a regularly sloped demand (elastic), that is, when $a > 0$ and $b = 0$.

- (c) Observe in particular what happens to the dead-weight loss in each of these extreme cases. Conclude that whenever we have one of the extreme cases of either a perfectly inelastic demand or a perfectly inelastic supply, *the dead-weight loss is zero*.
- (d) Use the previous item your previous calculations to conclude that when we have one of these extreme cases, the side of the market that is perfectly inelastic pays the whole amount of the tax/receives the whole amount of the subsidy.

Solution

- (a) If $a = 0$ then $P^S = P^e$, which implies that the consumer pays the whole tax/receives the whole subsidy, as can be seen by the consumer incidence expression. Furthermore Q^T and Q^S are equal to Q^e .
- (b) If $b = 0$ then $P^d = P^e$, which implies that the producer pays the whole amount of the tax/receives the whole subsidy, as can be seen by the producer incidence expression. Furthermore, the dead-weight loss in this case is also zero. Furthermore Q^T and Q^S are equal to Q^e .
- (c) In both the tax and subsidy cases, the dead-weight loss expression becomes zero if either $a = 0$ or $b = 0$. When demand or supply are perfectly inelastic, the distortion created by the tax or subsidy is zero and therefore the dead-weight loss is zero.
- (d) As mentioned before, when demand is inelastic, the tax incidence/subsidy incidence expression says that the consumer will pay the whole tax/receive the whole subsidy. Similarly, when supply is inelastic, the tax incidence/subsidy incidence expression says that the producer will pay the whole amount of the tax/receive the whole subsidy.

4. Now we will analyze the case of perfectly elastic demand or perfectly elastic supply.

- (a) A perfectly elastic demand corresponds to the case in which any quantity is consumed at a specific price P^* . Think about a horizontal line at P^* when graphing this demand equation. Note that the efficient equilibrium price will be precisely P^* . When supply is regularly sloped but demand is perfectly elastic, the price buyers pay will be $P^d = P^*$. The price sellers will receive will be different from P^* and depends on whether we have a tax or a subsidy. Repeat the calculations that you have done in the previous parts. In particular, note that suppliers will pay the whole tax/receive the whole subsidy since they are the side that is more inelastic.
- (b) A perfectly elastic supply corresponds to the case in which any quantity is supplied at a specific price P^* . Think about a horizontal line at P^* when graphing this supply equation. Once again, note that the efficient equilibrium price will be P^* . When demand is regularly sloped but supply is perfectly elastic, the price sellers receive will be $P^S = P^*$. The price buyers will pay will be different from P^* and depends on whether we have a tax or a subsidy. Repeat the calculations that you have done in the previous parts. In particular, note that buyers will pay the whole tax/receive the whole subsidy since they are the side that is more inelastic.

Solution

- (a) Consider first the case of perfectly elastic demand. We have that $Q^e = B + bP^*$ and $P^e = P^*$.

	Tax	Subsidy
P^d	P^*	P^*
P^s	$P^* - T$	$P^* + S$
$Q^T Q^S$	$B + bP^* - bT = Q^e - bT$	$B + bP^* + bS = Q^e + bS$
Revenue Expenditure	$T(B + bP^* - bT)$	$S(B + bP^* + bS)$
Consumer Incidence Producer Incidence	0 1	0 1
Dead-Weight Loss	$\frac{bT^2}{2}$	$\frac{bS^2}{2}$

- (b) Now consider the case of perfectly elastic supply. We have that $Q^e = A - aP^*$ and $P^e = P^*$.

	Tax	Subsidy
P^d	$P^* + T$	$P^* - S$
P^s	P^*	P^*
$Q^T Q^S$	$A - aP^* + -T = Q^e - aT$	$A - aP^* + aS = Q^e + aS$
Revenue Expenditure	$T(A - aP^* - aT)$	$S(A - aP^* + aS)$
Consumer Incidence Producer Incidence	1 0	1 0
Dead-Weight Loss	$\frac{aT^2}{2}$	$\frac{aS^2}{2}$

5. Consider the following scenarios¹: (demand and supply equations with one tax and one subsidy)

- $Q = 100 - 2P^d$, $Q = 10 + P^s$, a tax of $T = 10$, a subsidy of $S = 10$.
- $Q = 10 - P^d$, $Q = -\frac{2}{3} + \frac{P^s}{3}$, a tax of $T = 4$, a subsidy of $S = 3$.
- $Q = 95,000 - 50P^d$, supply is fixed at $Q = 90,000$, a tax of $T = 50$, a subsidy of $S = 20$.
- $Q = 600 - P^d$, $Q = P^s$, a tax of $T = 300$, a subsidy of $S = 400$.
- $Q = 28 - P^d$, supply is $P = 20$, a tax of $T = 1$, a subsidy of $S = 10$.

For each one of these cases, compute

- Amount of tax paid/amount of subsidy received by consumers.
- Amount of tax paid/amount of subsidy received by sellers.
- Government revenue from the tax/Government expenditure from the subsidy.
- Consumer surplus before and after the tax/subsidy.
- Producer surplus before and after the tax/subsidy.
- Dead-weight loss from the tax/subsidy.
- Define the following variables:

CS = consumer surplus
 PS = producer surplus
 $SS = CS + PS$ = social surplus
 R = Government revenue from the tax
 E = Government expenditure from the subsidy
 DWL = Dead-weight loss

Observe carefully the calculations you made in each of the cases in this exercise. Use them to conclude that the following identities hold:

$$\begin{aligned} \text{For a tax: } SS^{\text{before tax}} &= SS^{\text{after tax}} + R + DWL \\ \text{For a subsidy: } SS^{\text{after subsidy}} + DWL &= SS^{\text{before subsidy}} + E \end{aligned}$$

Use these identities to conclude that:

- Whenever there is a tax, part of the original social surplus is lost, that is, neither society nor the government appropriates it.
- Whenever there is a subsidy, government expenditures are covering the change in surplus to society plus an extra amount that nobody appropriates and thus is lost.

Solution

Table 2 Provides the calculations for taxes while table 3 shows the corresponding calculations for subsidies.

From both tables you can check that the identities indeed hold. Consider the following:

- For a tax, where $SS^{\text{before tax}} \geq SS^{\text{after tax}}$, the change in social surplus is $\Delta SS = SS^{\text{before tax}} - SS^{\text{after tax}}$. Thus, using the identity, we can write

$$\Delta SS - DWL = R,$$

¹I assigned cases ii-v in assignment 5, part 2. Case i is the one I solved in lecture 6, part 2.

which shows that government revenues are given by the change in social surplus minus the dead-weight loss which shows that indeed some part of the surplus is lost irrevocably.

- For a subsidy, where $SS^{\text{before subsidy}} \leq SS^{\text{after subsidy}}$, the change in social surplus is $\Delta SS = SS^{\text{after subsidy}} - SS^{\text{before subsidy}}$. Thus, using the identity, we can write

$$\Delta SS + DWL = E,$$

which shows that resources from government expenditures are used to fund the change in social surplus but part of these resources are also being lost in the form of an irrevocable loss. ν

Table 2: Calculations for taxes

	Case I	Case II	Case III	Case IV	Case V
Amount of tax paid by consumers	111.11	1.00	0.00	22,500.00	7.00
Amount of tax paid by producers	222.22	3.00	4,500,000.00	22,500.00	0.00
Government revenue from the tax	333.33	4.00	4,500,000.00	45,000.00	7.00
Consumer surplus before tax	400.00	2.00	81,000,000.00	45,000.00	32.00
Producer surplus before tax	750.00	6.00	9,000,000.00	45,000.00	0.00
Social surplus before tax	1,150.00	8.00	90,000,000.00	90,000.00	32.00
Consumer surplus after tax	277.78	0.50	81,000,000.00	11,250.00	24.50
Producer surplus after tax	505.56	1.50	4,500,000.00	11,250.00	0.00
Social surplus after tax	783.33	2.00	85,500,000.00	22,500.00	24.50
Dead Weight Loss	33.33	2.00	0.00	22,500.00	0.50

Table 3: Calculations for subsidies

	Case I	Case II	Case III	Case IV	Case V
Amount of subsidy received by consumers	155.56	2.06	0.00	100,000.00	180.00
Amount of subsidy received by producers	311.11	6.19	1,800,000.00	100,000.00	0.00
Government expenditure from the subsidy	466.67	8.25	1,800,000.00	200,000.00	180.00
Consumer surplus before subsidy	400.00	2.00	81,000,000.00	45,000.00	32.00
Producer surplus before subsidy	750.00	6.00	9,000,000.00	45,000.00	0.00
Social surplus before subsidy	1,150.00	8.00	90,000,000.00	90,000.00	32.00
Consumer surplus after subsidy	544.44	3.78	81,000,000.00	125,000.00	162.00
Producer surplus after subsidy	1,038.89	11.34	10,800,000.00	125,000.00	0.00
Social surplus after subsidy	1,583.33	15.13	91,800,000.00	250,000.00	162.00
Dead Weight Loss	33.33	1.13	0.00	40,000.00	50.00